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Power supply system

Abstract

A power supply system for a vehicle for controlling power supply from a second power supply provided for backup of the first power supply to a plurality of in-vehicle equipment, comprising a relay inserted between the second power supply and the first in-vehicle equipment, when a collision of the vehicle is detected during the backup operation of the first power supply by the second power supply, the electric power supplied from the second power supply to the second in-vehicle equipment, characterized in that it supplies in parallel to the first in-vehicle equipment via the relay.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2023-102235 filed on Jun. 22, 2023, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a power supply system for a vehicle that controls electric power supply from a sub-power supply provided for backup of a main power supply, to a plurality of pieces of in-vehicle equipment.

2. Description of Related Art

(3) Japanese Patent No. 5618024 discloses a backup power supply device that has a capacitor as a sub-battery, a charging circuit provided on a charging path of the capacitor and performing a step-down operation, a step-up circuit provided on an output path of the capacitor, and a door lock release output terminal connected to the step-up circuit. This backup power supply device described in U.S. Pat. No. 5,618,024 performs backup of a door lock (mechanical door) system in which a door is locked/unlocked using a motor in an emergency such as when a main battery is damaged due to a traffic accident or the like, using the capacitor.

SUMMARY

(4) In recent years, there are vehicles in which both the above door lock system and an “e-Latch” system, for locking/unlocking a door using a latch mechanism, are installed. In a vehicle equipped with both the door lock system and the e-Latch system in this way, when attempting to realize backup processing of each system, door unlock control of these systems differ from each other. Accordingly, separate sub-batteries are necessary for backup. Thus, attempts to integrate backup functions of the two systems into one so that only one sub-battery is necessary are not easy, since there is a possibility that changes will need to be made to the control (complication of control), the number of parts other than the sub-battery will increase, and so forth.

(5) In this way, there is room for further study regarding a power supply system capable of performing backup processing of both the door lock system and the e-Latch system in an emergency, in terms of increased size, increased number of parts, and increased costs.

(6) The present disclosure has been made in view of the above problem, and an object thereof is to provide a power supply system capable of suppressing increase in size, increase in the number of parts, increase in costs, and so forth.

(7) In order to solve the above problem, according to an aspect of the technique of the present disclosure, a power supply system for a vehicle, for controlling electric power supply from a second power supply provided for backup of a first power supply to a plurality of pieces of in-vehicle equipment, includes a relay interposed between the second power supply and first in-vehicle equipment. When collision of the vehicle is detected during a backup operation of the first power supply by the second power supply, electric power supplied from the second power supply to second in-vehicle equipment is supplied in parallel to the first in-vehicle equipment via the relay.

(8) According to the power supply system of the present disclosure, increase in size, increase in the number of parts, increase in costs, and so forth, can be suppressed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

(2) FIG. 1 is a functional block diagram of a power supply system and a peripheral portion thereof according to an embodiment of the present disclosure;

(3) FIG. 2 is a process flow chart of power-supply control executed by a power supply system; and

(4) FIG. 3 is a diagram illustrating a configuration of a power supply system according to an embodiment of the present.

DETAILED DESCRIPTION OF EMBODIMENTS

(5) The power supply system according to the present disclosure includes a relay between the sub-power supply and the mechanical door system, and supplies electric power supplied from the sub-power supply to the E latch system in parallel to the mechanical door system through the relay when a collision of the vehicle is detected during the backup operation of the main power supply by the sub-power supply. Thus, both the mechanical door system and the E latch system can be supplied with backup power with a single output from one sub-power supply (integration of the backup function). Hereinafter, an embodiment of the present disclosure will be described in detail with reference to the drawings.

EMBODIMENT

(6) Configuration

(7) FIG. 1 is a functional block diagram of a power supply system **100** and a peripheral portion thereof according to an embodiment of the present disclosure. The functional block illustrated in FIG. 1 includes a power supply system **100**, a plurality of systems including an SBW **210**, a body ECU **220**, a mechanical door **230**, an E-latch **240**, and a brake **250**, and a first power supply **300**. The power supply system **100**, SBW **210**, the body ECU **220**, the mechanical door **230**, the E-latch **240**, the brake **250**, and the first power supply **300** may be mounted on vehicles.

(8) The first power supply **300** provides power to a plurality of systems, such as the SBW **210**, the body ECU **220**, the mechanical door **230**, the E-latch **240**, and the brake **250**. The first power supply **300** is a power supply (+B power supply). Examples of the first power supply **300** include a generator including an alternator that generates electric power and a DCDC converter that converts the generated electric power into a predetermined voltage, and a predetermined power supply configuration such as a rechargeable battery that is configured to be chargeable and dischargeable such as a lithium-ion battery or a lead-acid battery.

(9) A plurality of systems such as an SBW **210**, a body ECU **220**, a mechanical door **230**, an E-latch **240**, and a brake **250** are in-vehicle equipment for realizing predetermined functions related to vehicles. These may be systems that require a particularly redundant power supply configuration. The plurality of systems is connected and configured such that electric power can be supplied from the first power supply **300** without the power supply system **100**, and backup power based on the first power supply **300** can be supplied from the power supply system **100**. Note that the plurality of systems mounted on the vehicle are not limited to those illustrated in FIG. 1.

(10) SBW **210** is a shift-by-wire control system that can change the gear stage of a transmission (not shown) by an electric signal. SBW **210** includes an SBW-ECU that is an electronic control unit that converts a shift operation of a driver into an electric signal, and an SBW-ACT that is an actuator that changes a gear stage based on an electric signal instructed from SBW-ECU. SBW **210** is connected to an output (A output) of SW **115** so as to be able to supply power from an integrated sub-battery **110** described later.

(11) The body ECU **220** is one of electronic control units that are equipment mounted on a vehicle, and performs control related to a body system of a so-called vehicle. The body ECU **220** according to the present embodiment can detect that a collision of a vehicle has occurred, and when a collision of the vehicle has been detected, a predetermined TRG is outputted to the power supply system **100**. TRG signal may be, for example, a signal that is High if a collision of the vehicle is not detected and a signal that is Low if a collision of the vehicle is detected. The body ECU **220** is connected to an output (B1 output) of SW **116** so as to be able to supply power from an integrated sub-battery **110** described later.

(12) The mechanical door **230** is a system that performs door lock control capable of locking/unlocking a door of a vehicle. The mechanical door **230** includes a door unlock ECU, a door unlock ACT, and the like. The door unlock ECU is an electronic control unit that converts a locking and unlocking operation of the driver into an electric signal. The door unlock ACT is a motor actuator that changes a locked state and an unlocked state of a door of a vehicle based on an

electric signal instructed from a door unlock ECU. As the mechanical door **230**, a vehicle speed sensitive automatic door system can be exemplified.

(13) Like the mechanical door **230**, the E latch **240** is another system that performs door lock control capable of locking/unlocking the door of the vehicle. The E latch **240** has an electric latch mechanism that does not require power supply switching by a mechanical relay for motor driving. The E-latch **240** is connected to an output (B2 output) of SW **117** so as to be able to supply power from an integrated sub-battery **110**, which will be described later.

(14) The brake **250** is a system that performs brake control capable of generating a braking force in the vehicle. The brake **250** includes a brake ECU that is an electronic control unit that converts a brake operation of a driver into an electric signal, and a brake ACT that is an actuator that generates a braking force based on an electric signal instructed from the brake ECU. The brake **250** is connected to an output (C output) of SW **118** so as to be able to supply power from an integrated sub-battery **110** described later.

(15) The power supply system **100** is configured to function as a sub-power supply to back up electric power to SBW **210**, body ECU **220**, mechanism door **230**, E-latch **240**, and brake **250** when power to SBW **210**, body ECU **220**, mechanism door **230**, E-latch **240**, and brake **250** from the first power supply **300** is abnormal, such as due to a power failure of the first power supply **300**, which is the main power supply.

(16) The power supply system **100** illustrated in FIG. **1** includes an integrated sub-battery **110** including a second power supply **111**, a DDC **112**, a collision determination unit **113**, a control unit **114**, and a plurality of SW **115 118**, a relay **121**, and a diode **122**.

(17) The second power supply **111** is, for example, a power supply source constituted by a secondary battery such as a lithium-ion battery configured to be chargeable and dischargeable, or a power storage element such as a capacitor. The second power supply **111** is provided as a sub power supply source for backing up the first power supply **300**, which is a main power supply source, in an emergency or the like. The second power supply **111** is connected to DDC **112** so that the electric power of the first power supply **300** can be charged and the electric power (back-up electric power) stored therein can be discharged to SBW **210**, the body ECU **220**, the mechanical door **230**, the E-latch **240**, and the brake **250**.

(18) DDC **112** is a DCDC converter (step-up/step-down type) for charging and discharging the electric power of the second power supply **111**. In this DDC **112**, the electric power inputted from the first power supply **300** can be converted into electric power of a predetermined voltage and outputted to the second power supply **111** based on an instruction (such as a voltage command value) from the control unit **114**. DDC **112** may convert the electric power stored in the second power supply **111** into electric power of a predetermined voltage and may be outputted from the plurality of SW **115** to SBW **210**, the body ECU **220**, the mechanical door **230**, the E-latch **240**, and the brake **250** via the **118**.

(19) The collision determination unit **113** is configured to receive a TRG signal from the body ECU **220** and determine whether or not a collision of vehicles has occurred based on TRG signal. Specifically, the collision determination unit **113** determines that a collision of the vehicle has occurred when a Low level TRG signal is inputted. When the collision determination unit **113** determines that there is a collision of the vehicle, information related to the determination is notified to the control unit **114**.

(20) The control unit **114** is configured to control the operation of DDC **112** and control the switching of the plurality of SW **115 118** based on the power supply status of the first power supply **300** and the determination information inputted from the collision determination unit **113**. The control unit **114** may acquire the power supply status of the first power supply **300** directly from the first power supply **300** or indirectly from an ECU (not shown) that monitors/controls the first power supply **300**. When a power supply failure of the first power supply **300** occurs due to some reason (for example, an abnormality of the first power supply **300**, a disconnection of a wire, a

ground fault, or the like) and when the collision determination unit **113** determines that there is a collision of vehicles, the control unit **114** appropriately switches the connected state (conduction/disconnection) of the plurality of SW **115 118**. The control unit **114** suitably controls backup power supply from the second power supply **111** to SBW **210**, the body ECU **220**, the mechanical door **230**, the E-latch **240**, and the brake **250**. Details of the backup power supply control by the control unit **114** will be described later.

(21) The plurality of SW **115 118** are configured to switch the power supply status (supply/non-supply) from the second power supply **111** to SBW **210**, the body ECU **220**, the mechanical door **230**, the E-latch **240**, and the brake **250**. SW **115** is provided between DDC **112** and SBW **210**. SW **116** is provided between DDC **112** and the body ECU **220**. SW **117** is provided between DDC **112** and the E-latch **240**. A SW **118** is provided between DDC **112** and the brake **250**. As SW **115 118**, a switch device such as a mechanical switch or a semi-conductor switch that operates based on an instruction from the control unit **114** is used.

(22) The relay **121** is a configuration for controlling backup power supply to the mechanical door **230**. The relay **121** is provided between the integrated sub-battery **110** and the mechanical door **230**. The relay **121** is, for example, a single-pole, single-throw excitation type mechanical relay. In the excitation mechanical relay, the coil portion is connected between the output of SW **116** (B1 output) and the wire through which TRG signal output from the body ECU **220** flows, and the contact portion is connected between the output of SW **117** (B2 output) and the mechanical door **230** (power supply line). In the configuration illustrated in FIG. **1**, B1 power of SW **116** for the body ECU **220** is used as the power supply of the coil portion of the relay **121**. However, as in the configuration shown in FIG. **3**, B2 power of SW **117** for the E-latch **240** may be used for the power supply of the coiled portion of the relay **121**. With this connection, when backup is required, a configuration is constructed in which power supply is supplied directly from the integrated sub-battery **110** to the E latch **240**, and power is supplied from the integrated sub-battery **110** to the mechanical door **230** via the relay **121**.

(23) The diode **122** is inserted between the coil end of the relay **121** and the wiring of TRG signal with the direction from the coil end to the wiring of TRG signal as the forward direction. The diode **122** is provided to suppress an unintended current from flowing through the coil portion of the relay **121** (current backflow from TRG signal to B1 output) to SW **116** or the body ECU **220** when TRG signal is at High level (in a normal state in which there is no collision detection). The diode **122** can prevent a fault detection malfunction of the integrated sub-battery **110** caused by the current backflow.

(24) Control

(25) Next, the operation of the power supply system **100** according to an embodiment of the present disclosure will be described with further reference to FIG. **2**. FIG. **2** is a flowchart illustrating a processing procedure of backup power supply control executed by the power supply system **100**. The backup power supplying control illustrated in FIG. **2** is started, for example, when the ignition relay of the vehicle is turned IGR-ON.

(26) S201

(27) The power supply system **100** completes the necessary charging of the second power supply **111**. When the power supply of the first power supply **300** fails due to the completion of the charge, SBW **210** using the second power supply **111**, the body ECU **220**, the mechanical door **230**, the E-latch **240**, and the brake **250** can be supplied with backup power. When the required charge to the second power supply **111** is completed, the process proceeds to S202.

(28) S202

(29) The power supply system **100** determines whether a power failure of the first power supply **300** has occurred. The power failure of the first power supply **300** includes, for example, a condition in which electric power cannot be normally supplied from the first power supply **300** to SBW **210**, the body ECU **220**, the mechanical door **230**, the E-latch **240**, and the brake **250**, such as

an abnormality or a failure of the first power supply **300**, a disconnection of the power supply line, and a ground fault. If the power supply system **100** determines that the power supply failure of the first power supply **300** has occurred (S202, Yes), the process proceeds to S203. On the other hand, when the power supply system **100** determines that the power supply failure of the first power supply **300** has not occurred (S202, No), the backup power supply control is ended.

(30) S203

(31) The power supply system **100** switches the status of SW **115 118** by the control unit **114** so that backup electric power can be supplied from the integrated sub-battery **110** to SBW **210**, the body ECU **220**, and the brake **250**. Specifically, the power supply system **100** controls SW **115** to be in a conductive state to turn on the supply of the A output, controls SW **116** to be in a conductive state to turn on the supply of B1 output, controls SW **117** to be in a cutoff state to turn off the supply of B2 output, and controls SW **118** to be in a conductive state to turn on the supply of the C output. When SW **115** controls SBW **210**, the body ECU **220**, and the **118** so that back-up power can be supplied to the brake **250**, the process proceeds to S204.

(32) S204

(33) The power supply system **100** determines whether or not the collision determination unit **113** determines that a vehicle collision has occurred. This determination is made by detecting Low of TRG that changes in response to the body ECU **220** detecting a crash of the vehicle. If it is determined that a vehicle-collision has occurred (S204, Yes), the process proceeds to S205. On the other hand, if it is not determined that the vehicle has collided (S204, No), the backup power supply control is terminated while the supply of B2 power remains off.

(34) S205

(35) The power supply system **100** controls SW **117** to be conductive by the control unit **114** to switch B2 power from the supply-off state to the supply-on state. When B2 power is controlled to be supplied, the process proceeds to S206.

(36) S206

(37) The power supply system **100** is in a state in which backup electric power can be supplied from the integrated sub-battery **110** to the E-latch **240** by switching SW **117** to the conductive state (backup power supply). When the back-up power supply to the E-latch **240** is enabled, the process proceeds to S207.

(38) S207

(39) The power supply system **100** starts supplying power from the integrated sub-battery **110** via the relay **121** by controlling B2 power to supply on (relay supply on). In the above-described S204, TRG signal is Low in response to the body ECU **220** detecting the collision of the vehicle. The excitation current flows through the coil-shaped portion of the relay **121**, and the contact portion is already in conduction. Therefore, when B2 power is turned on, the relay supply transitions to the ON state accordingly. When the relay-supply is turned on, the process proceeds to S208.

(40) S208

(41) The power supply system **100** enables backup power supply from the integrated sub-battery **110** to the mechanical door **230** by turning on the relay supply. When the relay **121** is controlled in a state in which backup power supply to the mechanical door **230** is possible, this backup power supply control is ended.

(42) Operations and Effects

(43) As described above, according to the power supply system (power supply system **100**) according to an embodiment of the present disclosure, the relay circuit (relay **121**, diode **122**) is provided between the sub power supply source (second power supply **111**) provided for backup of the main power supply source (first power supply **300**) and the predetermined first in-vehicle equipment (mechanical door **230**). When a collision of the vehicle is detected during the backup operation of the main power supply source by the sub power supply source, the power supply system of the present embodiment performs control to supply the electric power supplied from the

sub power supply source to the predetermined second in-vehicle equipment (E latch **240**) in parallel to the first in-vehicle equipment via the relay circuit.

(44) By the control using the relay circuit, it is possible to integrate the two backup functions of the first in-vehicle equipment and the second in-vehicle equipment into one by using one sub-battery without providing a sub-battery necessary for backup for each of the first in-vehicle equipment and the second in-vehicle equipment having different system control. Therefore, it is expected that the size of the power supply system **100** may be reduced, the number of components may be increased, and the system cost may be increased.

(45) Although an embodiment of the present disclosure has been described above, the present disclosure can be regarded as not only a power supply system but also a control method performed by the power supply system, a program of the control method, a computer-readable non-transitory storage medium storing the program, a vehicle including the power supply system, and the like.

(46) The power supply system of the present disclosure can be used as a system for supplying backup electric power to a plurality of in-vehicle equipment such as an SBW/body/door unlock (mechanical door, E-latch)/brake using an integrated sub-battery.

Claims

1. A power supply system for a vehicle, for controlling electric power supply from a second power supply provided for backup of a first power supply to a plurality of pieces of in-vehicle equipment, the power supply system comprising a relay interposed between the second power supply and first in-vehicle equipment, wherein, when collision of the vehicle is detected during a backup operation of the first power supply by the second power supply, electric power supplied from the second power supply to second in-vehicle equipment is supplied in parallel to the first in-vehicle equipment via the relay.
 2. The power supply system according to claim 1, wherein the relay is an excitation mechanical relay, and supplies electric power from the second power supply to the first in-vehicle equipment by controlling a contact to a conducting state by supplying an excitation current to a coil portion in response to collision of the vehicle being detected.
 3. The power supply system according to claim 1, wherein: the first in-vehicle equipment is equipment that controls locking and unlocking of a door of the vehicle by a motor actuator; and the second in-vehicle equipment is equipment that controls locking and unlocking of a door of the vehicle by a latch mechanism.
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